

Script05: ANOVA stats

Sara Beier

2024-11-06

Contents

1	Clean and setup working space	2
2	Load and transform input data	2
3	ANOVA statistics Incubation Day 5	3
3.1	Two-way ANOVAS with Tukey Posthoc testing	3
3.2	Create Boxplot Graphics	5
4	Growth curves	7
4.1	Mixed model ANOVA with Posthoc tests for pairwise comparisons	7
4.2	Create Plot Growth Curves	8

1 Clean and setup working space

```
rm(list = ls())
loaded_packages <- setdiff(loadedNamespaces(), c("base", "compiler", "datasets",
  "graphics", "grDevices", "grid", "methods", "parallel", "splines", "stats",
  "stats4", "tcltk", "tools", "utils"))
for (pkg in loaded_packages) {
  try(detach(paste0("package:", pkg), unload = TRUE, character.only = TRUE),
    silent = TRUE)
}

# Load packages
library(tidyverse)
library(car) #Levene's test
library(readxl) # read excel files (to read in Table S)
library(lme4) # mixed models via lmer
library(lmerTest) # mixed models via lmer + p-values
library(kableExtra)
library(scales)
library(ggtext)
library(emmeans)
# define functions Function for obtaining standard error, modified from
# (https://stackoverflow.com/questions/30485154/how-to-write-summaryse-function-in-r)
summarySE <- function(data = NULL, measurevar, groupvars = NULL, na.rm = FALSE,
  conf.interval = 0.95, .drop = TRUE) {
  library(plyr)
  length2 <- function(x, na.rm = FALSE) {
    if (na.rm)
      sum(!is.na(x)) else length(x)
  }
  dataac <- ddply(data, groupvars, .drop = .drop, .fun = function(xx, col) {
    c(N = length2(xx[[col]], na.rm = na.rm), mean = mean(xx[[col]], na.rm = na.rm),
      sd = sd(xx[[col]], na.rm = na.rm))
  }, measurevar)

  dataac <- rename(dataac, c(mean = measurevar))
  dataac$se <- dataac$sd/sqrt(dataac$N)
  return(dataac)
}
```

2 Load and transform input data

```
# Load data from Table S3
dat <- read_xlsx("../Data/Stables.xlsx", sheet = "Table_S3", .name_repair = "universal") %>%
  mutate_at(c(5:10), as.numeric)

dat$BGE <- dat$Bacterial.production/(dat$Bacterial.production + dat$Respiration)

# Calculate Bacterial Growth efficiency (BGE) Bulk production (ugCL.1d.1)
```

```
# need to be transformed to molar by the molecular weight of Carbon Bulk
# respiration (uM_h_corr) or umol L-1 h-1 need to be scaled to daily rate
```

```
# Transform data to improve match of residual to normal distribution
```

```
dat_trans <- dat %>%
  mutate(Cellcounts = log(Cellcounts)) %>%
  mutate(Bacterial.production = Bacterial.production) %>%
  mutate(Respiration = Respiration) %>%
  mutate(ASV.richness = 1/ASV.richness) %>%
  mutate(Shannon.entropy = log(Shannon.entropy)) %>%
  mutate(Inv.Simpson = Inv.Simpson) %>%
  mutate(BGE = BGE)
```

3 ANOVA statistics Incubation Day 5

3.1 Two-way ANOVAS with Tukey Posthoc testing

Two way ANOVAS with posthoc testing for pairwise comparisons are performed after inspecting normal distribution of residuals and testing for homogeneity of variances (levene's test)

```
par(mfrow = c(3, 3))
for (i in seq(5, dim(dat_trans)[2])) {
  dat.i <- dat_trans[dat_trans$Time == "T5", c(1:4, i)] %>%
    na.omit()
  x <- dat.i[, 5, drop = T]
  res.aov <- aov(x ~ Source.community * Medium, data = dat.i)
  qqnorm(resid(res.aov))
  legend("bottomright", c(names(dat_trans)[i]), , bty = "n")
}
```

```
stats.out <- data.frame()

for (i in seq(5, dim(dat_trans)[2])) {
  dat.i <- dat_trans[dat_trans$Time == "T5", c(1:4, i)] %>%
    na.omit()
  x <- dat.i[, 5, drop = T]
  levene.i <- leveneTest(x ~ Source.community * Medium, data = dat.i)[[3]][[1]]

  res.aov <- aov(x ~ Source.community * Medium, data = dat.i)

  stats.i <- data.frame(names = c("C-S", "CS-C", "CS-S"), p.adj = c(TukeyHSD(res.aov)[[1]][[11]],
    TukeyHSD(res.aov)[[1]][[10]], TukeyHSD(res.aov)[[1]][[12]]), diff = c(TukeyHSD(res.aov)[[1]][[2]],
    TukeyHSD(res.aov)[[1]][[1]], TukeyHSD(res.aov)[[1]][[3]]), test = rep(c(names(dat_trans)[i]),
    3), p.source = rep(summary(res.aov)[[1]][[5]][1], 3), F.source = rep(summary(res.aov)[[1]][[4]][
    3), p.DOM = rep(summary(res.aov)[[1]][[5]][2], 3), F.DOM = rep(summary(res.aov)[[1]][[4]][2],
    3), p.source.DOM = rep(summary(res.aov)[[1]][[5]][3], 3), F.source.DOM = rep(summary(res.aov)[[1]][[4]][
    3), levene = rep(levene.i, 3))

  stats.out <- rbind(stats.out, stats.i)
}
```

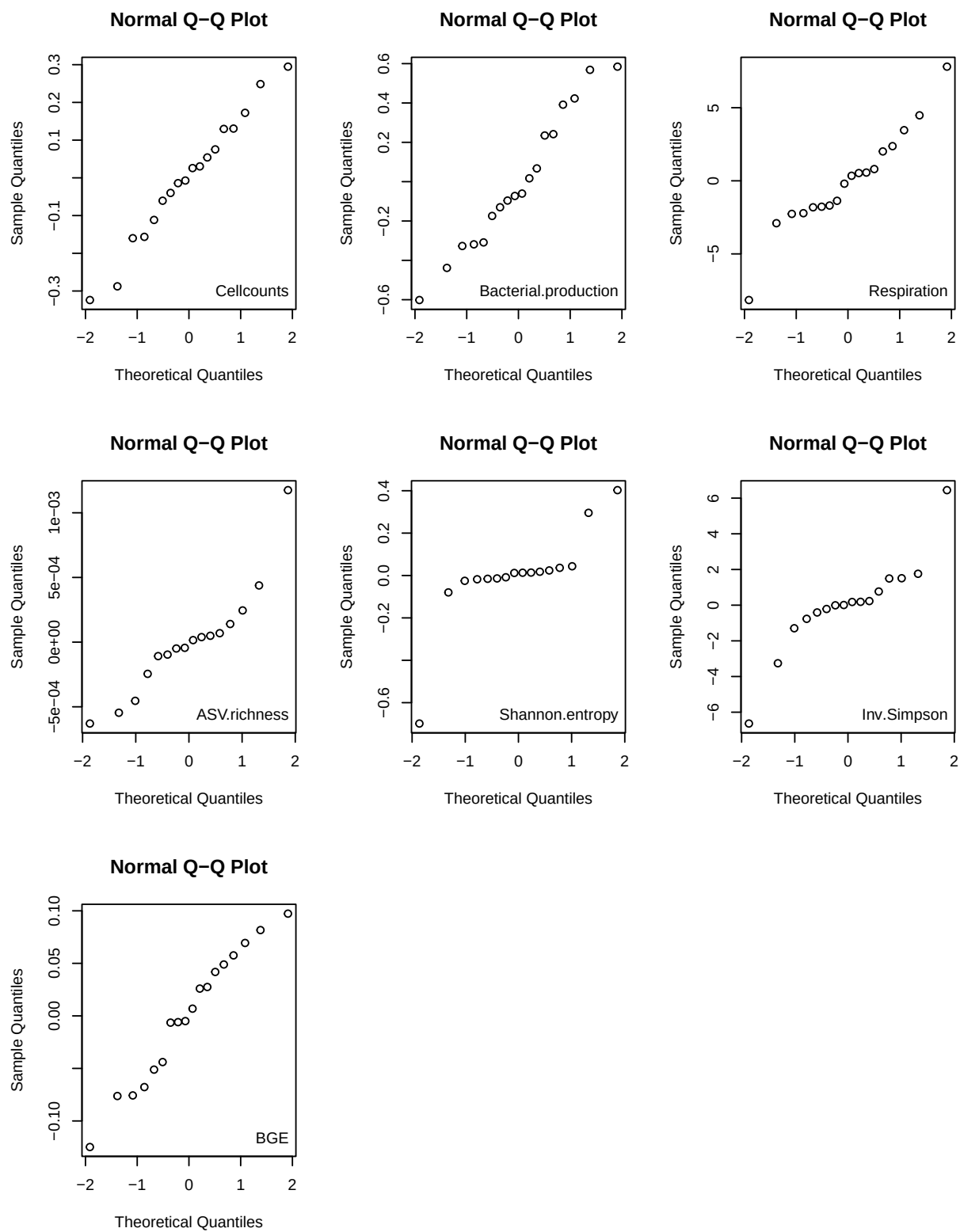


Figure 1: Inspect normality of residuals.

names	Tukey Posthoc		Two-way ANOVA							HOV
	p.adj	diff	test	p.source	F.source	p.DOM	F.DOM	p.source.DOM	F.source.DOM	levene
C-S	0.000	-0.861	Cellcounts	0.000	37.816	0.054	4.544	0.114	2.616	0.628
CS-C	0.992	0.014	Cellcounts	0.000	37.816	0.054	4.544	0.114	2.616	0.628
CS-S	0.000	-0.875	Cellcounts	0.000	37.816	0.054	4.544	0.114	2.616	0.628
C-S	0.015	-0.792	Bacterial.production	0.006	8.249	0.845	0.040	0.241	1.603	0.845
CS-C	0.933	0.085	Bacterial.production	0.006	8.249	0.845	0.040	0.241	1.603	0.845
CS-S	0.008	-0.876	Bacterial.production	0.006	8.249	0.845	0.040	0.241	1.603	0.845
C-S	0.957	-0.667	Respiration	0.010	6.854	0.417	0.708	0.756	0.287	0.489
CS-C	0.025	7.206	Respiration	0.010	6.854	0.417	0.708	0.756	0.287	0.489
CS-S	0.015	-7.873	Respiration	0.010	6.854	0.417	0.708	0.756	0.287	0.489
C-S	0.065	-0.001	ASV.richness	0.008	7.982	0.104	3.203	0.331	1.237	0.679
CS-C	0.008	-0.001	ASV.richness	0.008	7.982	0.104	3.203	0.331	1.237	0.679
CS-S	0.302	0.001	ASV.richness	0.008	7.982	0.104	3.203	0.331	1.237	0.679
C-S	0.179	-0.306	Shannon.entropy	0.149	2.318	0.655	0.212	0.779	0.256	0.478
CS-C	1.000	-0.005	Shannon.entropy	0.149	2.318	0.655	0.212	0.779	0.256	0.478
CS-S	0.250	-0.302	Shannon.entropy	0.149	2.318	0.655	0.212	0.779	0.256	0.478
C-S	0.035	-5.591	Inv.Simpson	0.042	4.443	0.433	0.667	0.755	0.289	0.243
CS-C	0.585	-2.142	Inv.Simpson	0.042	4.443	0.433	0.667	0.755	0.289	0.243
CS-S	0.277	-3.449	Inv.Simpson	0.042	4.443	0.433	0.667	0.755	0.289	0.243
C-S	0.276	-0.069	BGE	0.135	2.381	0.627	0.249	0.294	1.358	0.894
CS-C	0.136	-0.089	BGE	0.135	2.381	0.627	0.249	0.294	1.358	0.894
CS-S	0.891	0.020	BGE	0.135	2.381	0.627	0.249	0.294	1.358	0.894

```
kbl(stats.out, digits = 3) %>%
  kable_classic() %>%
  add_header_above(c(` ` = 1, ` Tukey Posthoc ` = 2, ` Two-way ANOVA ` = 7,
    HOV = 1)) %>%
  kable_styling(latex_options = "scale_down")
```

```
# print(format(stats.out,scientific = FALSE, digits = 6))
```

3.2 Create Boxplot Graphics

```
color=c('#7F95AA','#7F95AA','#7F95AA','darkolivegreen1','darkolivegreen1',
  'darkolivegreen1')
color2 = rep(c("black", "#CF5053", "black"),2)

dat_long <- dat %>%
  pivot_longer(
    cols = Cellcounts:BGE,
    names_to = "Variable",
    values_to = "Value"
  )
dat_long$Value[dat_long$Variable=="Cellcounts"]<-dat_long$Value[dat_long$Variable=="Cellcounts"]/100000

dat_long$Variable<-as.factor(dat_long$Variable)
dat_long$Variable<-factor(dat_long$Variable,c("Shannon.entropy","Inv.Simpson","ASV.richness",
  "Cellcounts","Bacterial.production", "Respiration", "BGE"))

# create a named vector of labels in the order you want
# my_labels <- c(
#   Shannon.entropy = "A) Shannon entropy",
#   Inv.Simpson = "B) Inv.Simpson",
```

```

# ASV.richness      = "C) ASV richness",
# Cellcounts       = "D) Cell densities",
# Bacterial.production = "E) Productivity",
# Respiration      = "F) Respiration",
# BGE              = "G) BGE"
# )
#
# my_labels2 <- c(
#   Shannon.entropy='Shannon-entropy',
#   Inv.Simpson='Inv.~Simpson',
#   ASV.richness='ASV~richness',
#   Cellcounts='atop(Cell~densities, 10^6~cells~mL^{-1})',
#   Bacterial.production='atop(Productivity, mu*mol~C~L^{-1}~day^{-1})',
#   Respiration='atop(Respiration, mu*mol~C~L^{-1}~day^{-1})',
#   BGE='BGE'
# )

my_labels_html <- c(
  Shannon.entropy="Shannon entropy",
  Inv.Simpson="Inv. Simpson",
  ASV.richness="ASV richness",
  Cellcounts="Cell densities<br>10<sup>6</sup> cells mL<sup>-1</sup>",
  Bacterial.production="Productivity<br>&mu;mol C L<sup>-1</sup> day<sup>-1</sup>",
  Respiration="Respiration<br>&mu;mol C L<sup>-1</sup> day<sup>-1</sup>",
  BGE="BGE"
)

base_ticks <- rep(c("Canet", "<span style='color:red;'>Coalesced</span>", "SOLA"), 2)

plot.panel<-dat_long[dat_long$Time=="T5",]%>%
  ggplot(aes(x=interaction(Source.community,Medium),y=Value,fill=Medium))+
  geom_boxplot(outlier.shape=NA)+
  #facet_wrap(~Variable,scales="free_y",labeller = labeller(Variable = my_labels))+
  facet_wrap(~Variable,scales="free_y",labeller = as_labeller(my_labels_html),
    strip.position="left")+

  geom_jitter(shape=21,width =0.15)+
  labs(x=NULL,y=NULL)+

  theme_bw()+theme(panel.grid.major = element_blank(),panel.grid.minor = element_blank(),
    strip.placement = "outside", # To place label outside axis ticks (y axis)
    strip.background = element_blank(),strip.text.x.top = element_text(hjust=0),
    strip.text.y.left = element_text(face = "bold", size = 9,lineheight = 0.2))+
  scale_fill_manual(values =c('#7F95AA','darkolivegreen1'),name=NULL)+
  theme(legend.position = c(0.5, 0), legend.text = element_text(size=14),
    legend.justification = c("left", "bottom"),
    plot.margin= margin(t = 5, r = 50, b = 5, l = 5),
    panel.spacing= unit(0.5, "lines")) +
  coord_cartesian(clip = "off")+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.3)))+
  scale_x_discrete(limits=levels(interaction(dat_long$Source.community,
    dat_long$Medium)),labels = base_ticks)+
  theme(axis.text.x = element_markdown(angle = 45, hjust=1))+

```

```

theme(strip.text.y.left = element_markdown(vjust = 0, lineheight = 0.9,
margin = margin(t=1, b=1, l=5, r=0)),strip.switch.pad.grid = uni
panel.spacing = unit(0.25, "lines"), # reduce inter-plot space
plot.margin = margin(t=5, b=10,l= 1, r=1))

```

```

egg::tag_facet(plot.panel,open="",tag_pool=c("A","B","C","D","E","F","G"),hjust=-0.1,vjust=1.15)

```

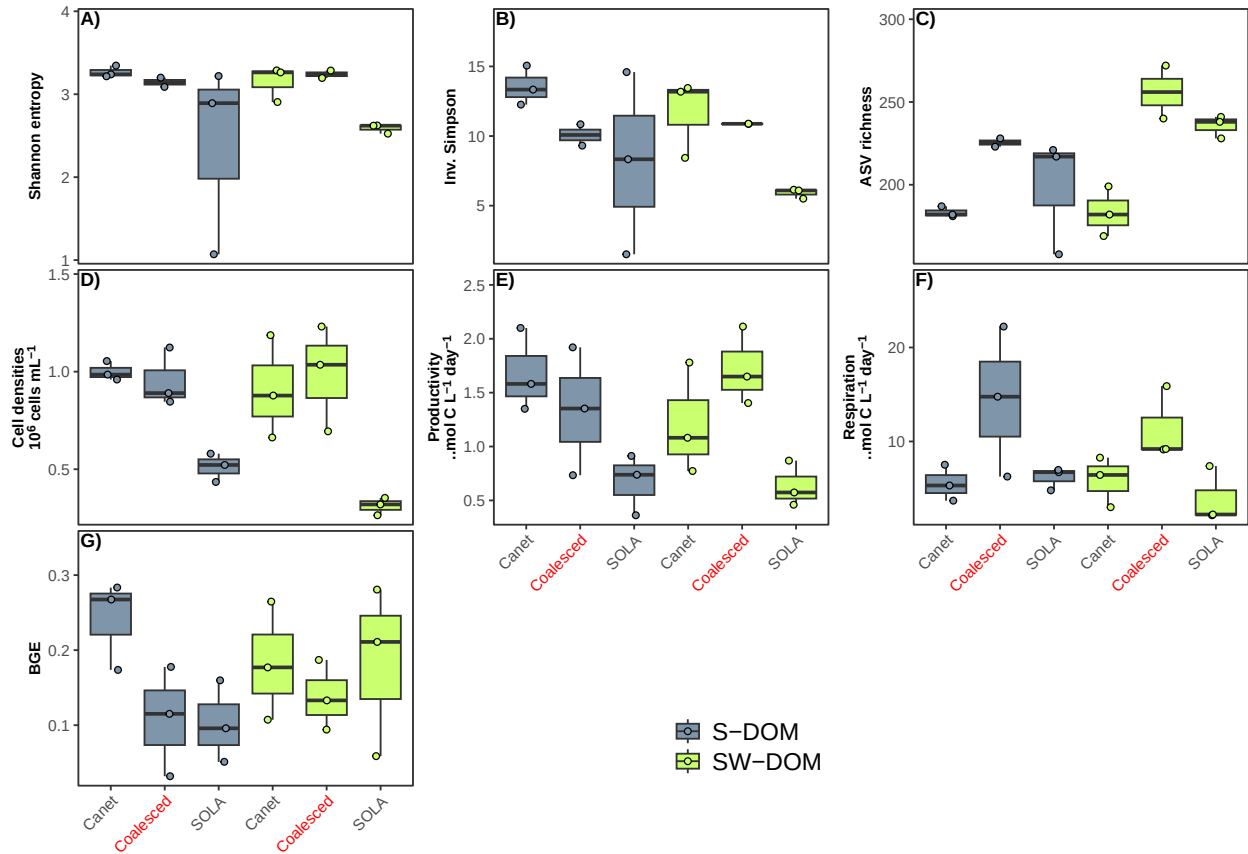


Figure 2: Figure 6: Boxplot figures illustration of diversity parameters (A-C) and functional characteristics (D-F) of experimental communities at day 5. Cell densities are given in 10⁶ cells/mL, productivity as well as respiration rates are given in μmol C/L day. Bacterial growth efficiency (BGE) is presented in proportion from 0-1. S-DOM and SW-DOM refer to media with DOM extracted from SOLA and SOLA+Warnow river water, respectively.

4 Growth curves

4.1 Mixed model ANOVA with Posthoc tests for pairwise comparisons

```

m <- lme4::lmer(log(Cellcounts) ~ Source.community + (1 | Time), data = dat)
qqnorm(resid(m))

```

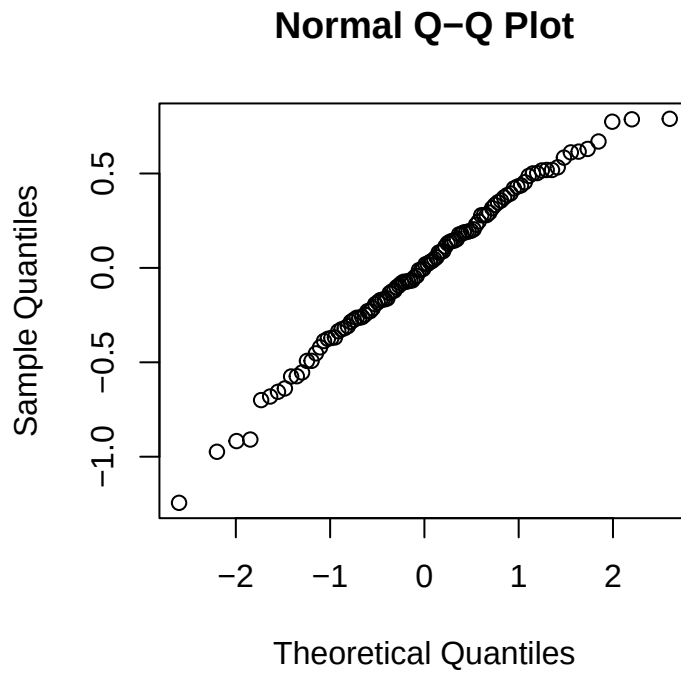


Figure 3: Inspect normality of residuals

```
emm.results <- emmeans(m, ~Source.community)
pairs(emm.results, adjust = "tukey")
```

```
## contrast estimate      SE df t.ratio p.value
## C - CS          0.241 0.0992 100   2.432  0.0440
## C - S           1.565 0.0992 100  15.771 <.0001
## CS - S          1.323 0.0992 100  13.339 <.0001
##
## Degrees-of-freedom method: kenward-roger
## Results are given on the log (not the response) scale.
## P value adjustment: tukey method for comparing a family of 3 estimates
```

4.2 Create Plot Growth Curves

```
## pdf
## 2
```

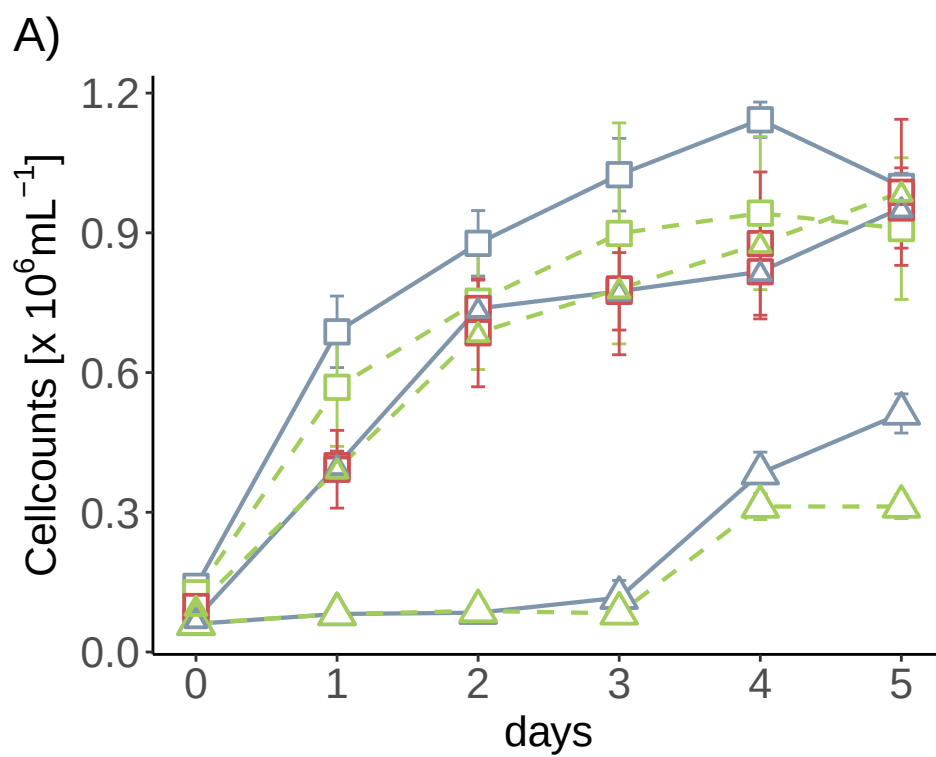


Figure 4: Growth curves of resuscitated communities (error bars indicate standard errors of triplicate incubations).